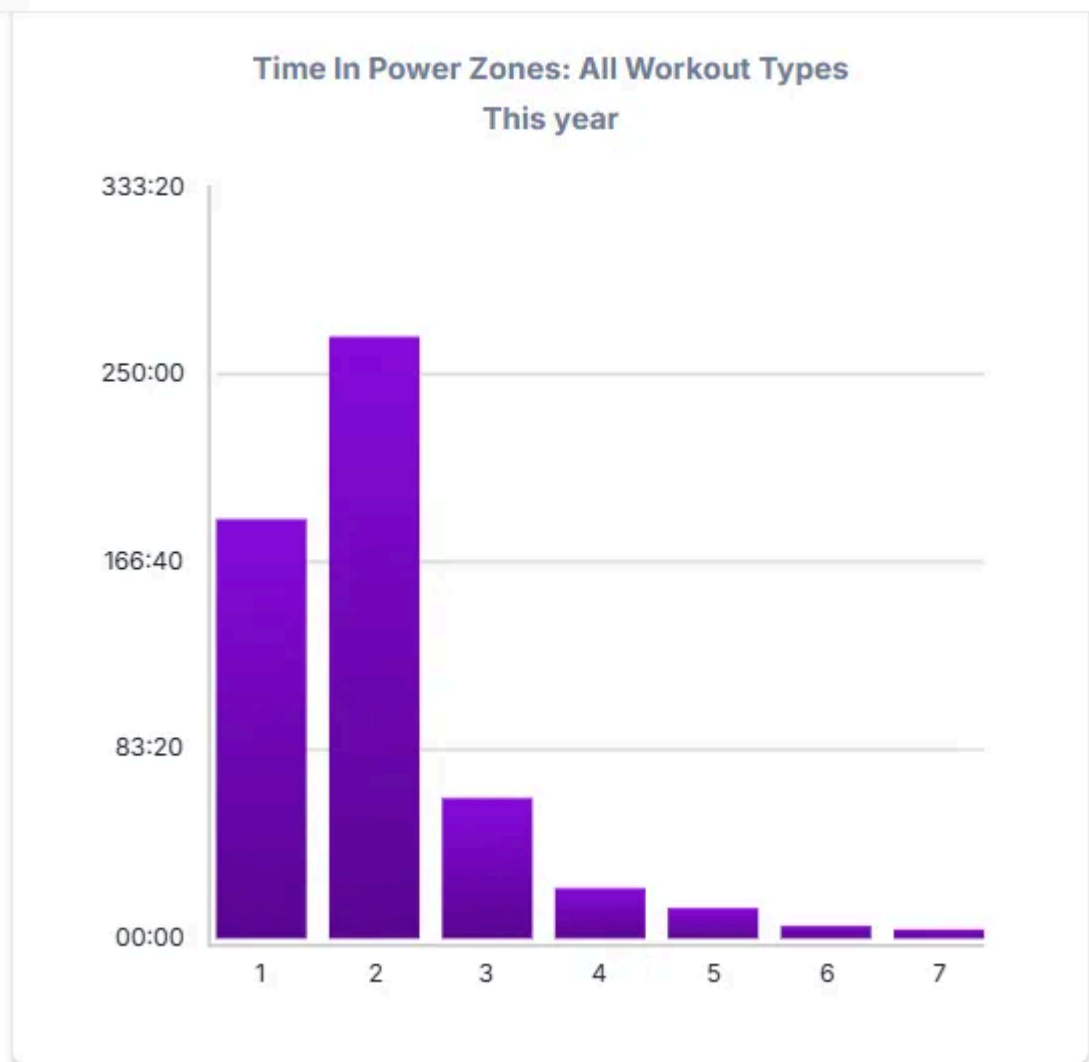


 knowledgeiswatt.substack.com/p/91-time-spent-in-intensity-zones

August 6, 2025

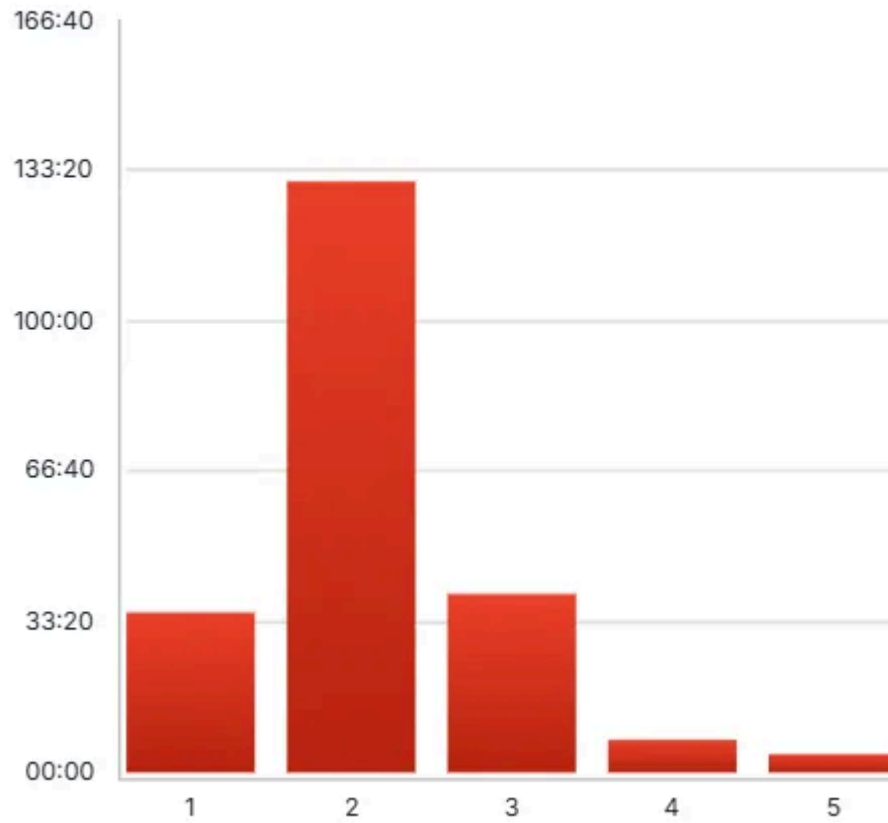


1/7



Screenshot from Training Peaks. Each number represents an intensity zone.

Time In Heart Rate Zones: All Workout Types



Screenshot from Training Peaks. Each number represents an intensity zone.

The two most used intensity distribution models by successful world-class endurance athletes are pyramidal and polarized:

- In the pyramidal model, most of the time is spent at low intensity, and a decreasing proportion of time at medium and high intensity.
- In the polarized model most of the time is spent at low intensity, but the amount of time spent at high intensity is more than the amount of time spent at medium intensity.

Polarized training seems to be the best solution to boost performance on short term (4-10 weeks). (1) On the other hand, when looking at the reported annual intensity distribution of some successful world-class road cyclists (2,3), they adopt a pyramidal intensity distribution with a switch from a less to a more polarized (even if still pyramidal) model when approaching the competition period. So, at the beginning of the periodization, they focus more on zone 3 (in a 5/7 zones model), while in the second part they reduce time spent in zone 3 (medium intensity) and increase the amount of work at high intensity (zone 4, 5, 6). The fact that polarized training is not adopted throughout all the season can be due to the fact that:

- performing high intensity training consistently for all the season can be too stressful (both physically and mentally)

- some data (4) also showed that doing the pyramidal-to-polarized progression seems also to be more effective from a pure physiological and performance point of view, when compared to maintaining a similar intensity distribution for all the year or doing the opposite switch (starting from a more polarized and progressing through a more pyramidal then).

The above cited intensity distribution model of world-class road cyclist was analysed using power output to describe the time spent in intensity zones. However, also heart rate can be used.

Can power and heart rate be usable interchangeably to evaluate the time spent in intensity zones, or the intensity distribution differs if described with power vs heart rate?

A study published in 2017 by a group led by Dajo Sanders (Newman University, Birmingham, UK - and World Tour Cycling Coach) on *International Journal of Sports Physiology and Performance* tried to answer this question. (5)

WHAT DID THEY DO?

- Fifteen male well-trained but not professional cyclists (VO2max: 55-69) competing at international and national level were recruited.
- First, they performed a step test in the lab to establish training zones using a 3-zones model:
 - Zone 1: below power/heart rate at first lactate threshold
 - Zone 2: power/heart rate between first and second lactate threshold
 - Zone 3: power/heart rate above the second lactate threshold.
- Then, they performed a 10 weeks training camp with each cyclists performing a personalized training plan prescribed by their own coach without any standardization between different athletes.

The goal was to see the agreement in time spent in intensity zones when using power vs heart rate.

WHAT DID THEY FIND?

The percentage time spent in zones resulted:

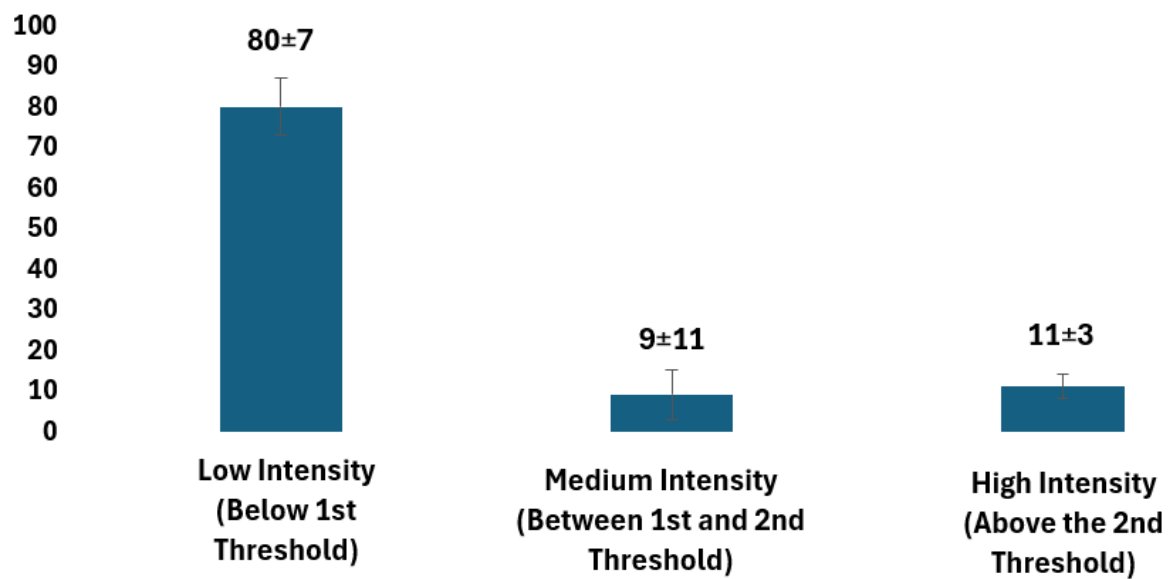
- **Polarized when using Power Zones:**

- Zone 1: $80 \pm 7\%$
- Zone 2: $9 \pm 6\%$
- Zone 3: $11 \pm 3\%$

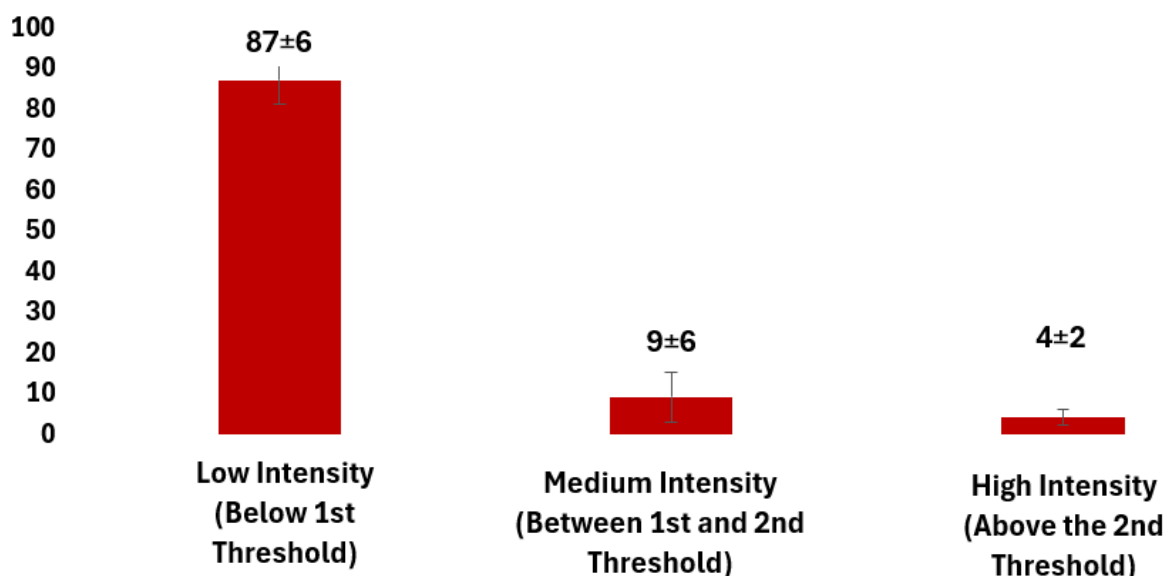
- **Pyramidal when using Heart Rate Zones:**

- Zone 1: $87 \pm 6\%$
- Zone 2: $9 \pm 6\%$
- Zone 3: $4 \pm 2\%$

**Time spent in Power
Intensity Zones
(%)**



Time spent in Heart Rate Intensity Zones (%)



Data from Sanders et al. (5)

PRACTICAL TAKEAWAYS:

- Intensity distribution/time spent in zones differs when expressed in power vs. heart rate.
- Specifically, heart rate intensity distribution underestimate time spent at high intensity when compared to using power output.

GABRIELE'S COMMENT

These data are very simple but still useful: I think they suggest us to not use power and heart rate intensity distribution interchangeably.

What is the reason behind the underestimation of time spent at high intensity when using heart rate?

The lower time spent at high intensity in heart rate vs. power zone is very likely due to the lag of heart rate response compared to power output when doing intervals. For example, when performing a two minutes' high intensity effort, it is only in the last minute that heart rate reaches the high intensity zone. Furthermore, when performing very short high intensity efforts (read sprints below 15s duration), heart rate doesn't even reach zone 2 in a 3 zones model. So, the mismatch between power and heart rate intensity zones is very likely to be larger when performing high intensity stochastic training sessions or races, especially criteriums, cyclocross and cross-country mountain bike events. Other scenarios which could increase the mismatch between heart rate and power time spent in zones are:

- altitude training: you can be at high/medium intensity when using heart rate but at medium/low intensity if looking at power.
- heat training/hot environment: same as altitude.

Thanks for Reading! Share with your Team!

Gabriele Gallo, PhD in Exercise and Sport Science

Founder of Knoweldgeiswatt

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